



Bahria University, Islamabad

Department of Software Engineering

Numerical Analysis Lab

(Fall-2026)

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Lab Journal: 01

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Error Basics: Rounding, Truncation & Significant Digits

Objectives

- Recognize and measure rounding errors in floating-point arithmetic.
- Understand truncation error when approximating infinite series.
- Understand the concept of significant digits.

Language: C++)

Task 1 – Rounding π

Write a program that:

- Stores π as a variable.
- Rounds it to 2, 4, and 6 decimal places.
- Prints the rounded values and the corresponding errors.

Code:

```
#include <iostream>
#include <cmath>    // Needed for round()
#include <iomanip>   // Needed for setprecision()

using namespace std;

int main() {

    double pi = 3.141592653589793;

    // 2. Rounding to 2, 4, and 6 decimal places using manual scaling
    // Formula: round(value * factor) / factor

    double pi_2 = round(pi * 100.0) / 100.0;        // 2 decimal places (10^2)
    double pi_4 = round(pi * 10000.0) / 10000.0;    // 4 decimal places (10^4)
    double pi_6 = round(pi * 1000000.0) / 1000000.0; // 6 decimal places (10^6)

    // Set precision for clean output formatting
    cout << fixed << setprecision(7);

    // 3. Print the rounded values and corresponding errors
    cout << "--- TASK 1: ROUNDING PI ---" << endl;
    cout << "Original Pi: " << pi << endl << endl;

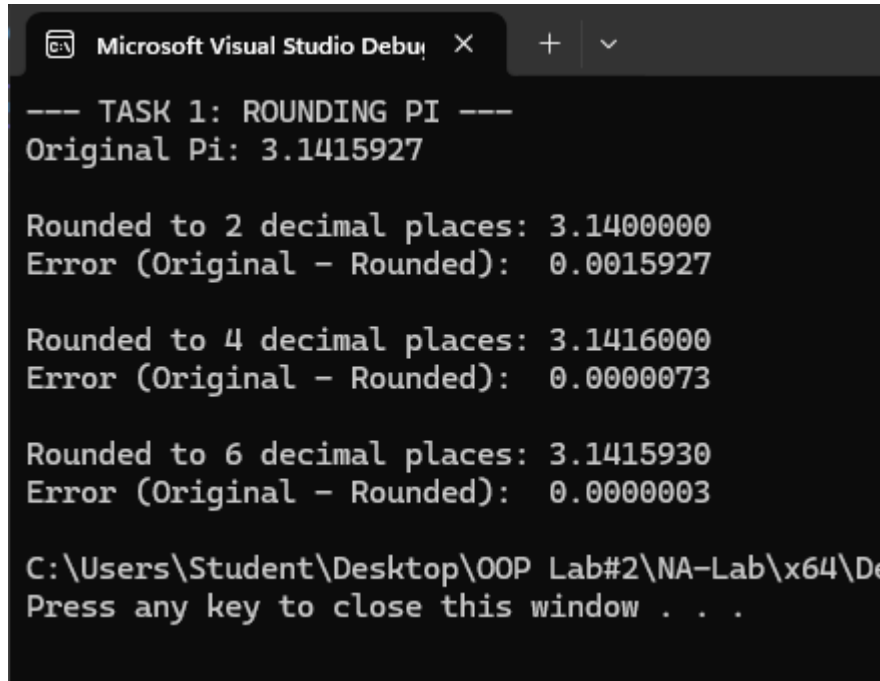
    cout << "Rounded to 2 decimal places: " << pi_2 << endl;
    cout << "Error (Original - Rounded): " << abs(pi - pi_2) << endl << endl;

    cout << "Rounded to 4 decimal places: " << pi_4 << endl;
    cout << "Error (Original - Rounded): " << abs(pi - pi_4) << endl << endl;

    cout << "Rounded to 6 decimal places: " << pi_6 << endl;
```

```
cout << "Error (Original - Rounded): " << abs(pi - pi_6) << endl;
    return 0;
}
```

Output:



```
Microsoft Visual Studio Debug Console
--- TASK 1: ROUNDING PI ---
Original Pi: 3.1415927

Rounded to 2 decimal places: 3.1400000
Error (Original - Rounded): 0.0015927

Rounded to 4 decimal places: 3.1416000
Error (Original - Rounded): 0.0000073

Rounded to 6 decimal places: 3.1415930
Error (Original - Rounded): 0.0000003

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Press any key to close this window . . .
```

Task 2 – Change the number of iterations

In the rounding-error program, instead of adding 0.1 one hundred times, change it to add 0.2 fifty times.

- What is the expected exact result?
- Does the error increase or decrease compared with the 0.1 case?

Code:

```
#include <iostream>
#include <iomanip> // Needed for setprecision()

using namespace std;

int main() {
    double x = 0.0;

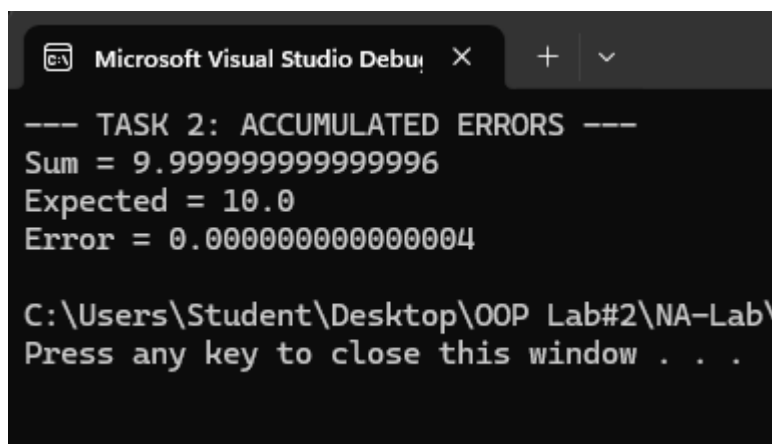
    // Task 2: Add 0.2 to x fifty times
    for (int i = 0; i < 50; i++) {
        x += 0.2;
    }
}
```

```
// Print with high precision to see the hidden rounding error
cout << fixed << setprecision(15);

cout << "--- TASK 2: ACCUMULATED ERRORS ---" << endl;
cout << "Sum = " << x << endl;
cout << "Expected = 10.0" << endl;
cout << "Error = " << 10.0 - x << endl;

return 0;
}
```

Output:



```
Microsoft Visual Studio Debug Console
--- TASK 2: ACCUMULATED ERRORS ---
Sum = 9.999999999999996
Expected = 10.0
Error = 0.000000000000004

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Press any key to close this window . . .
```

Task 3 – Extend the series Extend

the approximation to include the first 5 terms of the Taylor series and compare the error reduction.

Code:

```
#include <iostream>
#include <cmath>
#include <iomanip> // 1. Added this library to show more digits
using namespace std;

int main() {
    double x = 0.5;

    // Approximation using the first 5 terms
    double approx = x - pow(x, 3) / 6 + pow(x, 5) / 120 - pow(x, 7) / 5040 +
        pow(x, 9) / 362880;

    double trueVal = sin(x); // exact value
}
```

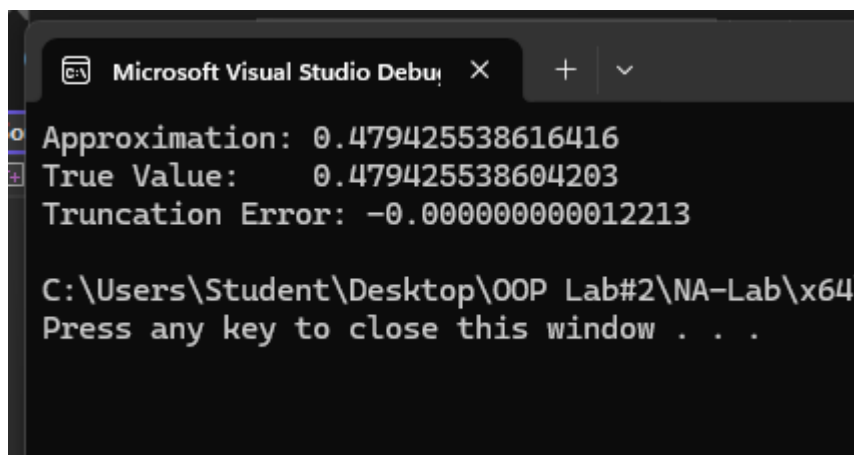
```
double error = trueVal - approx;

// 2. This line tells C++ to show 15 decimal places so you can see the error
cout << fixed << setprecision(15);

cout << "Approximation: " << approx << endl;
cout << "True Value:     " << trueVal << endl;
cout << "Truncation Error: " << error << endl;

return 0;
}
```

Output:



```
Microsoft Visual Studio Debug Console
Approximation: 0.479425538616416
True Value:    0.479425538604203
Truncation Error: -0.000000000012213

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Press any key to close this window . . .
```

Task 4 – Significant digits

Write a program that:

- Reads a number as a string.
- Counts how many significant digits it has.
- Prints the result.

Code:

```
#include <iostream>
#include <string>

using namespace std;

int main() {
    char choice;
```

```
do {
    string s;
    cout << "Enter a number: ";
    cin >> s;

    int n = s.length();
    int start = 0;
    int end = n - 1;
    bool hasDecimal = false;

    // Check if the number has a decimal point
    if (s.find('.') != string::npos) {
        hasDecimal = true;
    }

    // Skip leading zeros, signs (+/-), and decimal points
    while (start < n && (s[start] == '0' || s[start] == '+' || s[start] == '-' || s[start] == '.')) {
        start++;
    }

    // If the input is just zeros
    if (start == n) {
        cout << "Significant Digits: 0" << endl;
    }
    else {
        // If there is no decimal, ignore trailing zeros (e.g., 1200 -> 2 sig
figs)
        if (!hasDecimal) {
            while (end >= start && s[end] == '0') {
                end--;
            }
        }

        // Count the remaining valid digits
        int sigDigits = 0;
        for (int i = start; i <= end; i++) {
            if (s[i] != '.') {
                sigDigits++;
            }
        }

        cout << "Significant Digits: " << sigDigits << endl;

        // Ask the user if they want to check another number
        cout << "Do you want to check another number? (y/n): ";
        cin >> choice;
        cout << endl;

    } while (choice == 'y' || choice == 'Y');

    cout << "Program exited." << endl;

    return 0;
}
```

Output:

```
Microsoft Visual Studio Debug Console
Enter a number: 0.122
Significant Digits: 3
Do you want to check another number? (y/n): y

Enter a number: 0.022241
Significant Digits: 5
Do you want to check another number? (y/n): y

Enter a number: 12.0122
Significant Digits: 6
Do you want to check another number? (y/n): y

Enter a number: 0.00
Significant Digits: 0
Do you want to check another number? (y/n): y

Enter a number: 41.12000
Significant Digits: 7
Do you want to check another number? (y/n): n

Program exited.

C:\Users\Student\Desktop\OOP Lab#2\NA-Lab\x64\Debug\NA-Lab.exe
Press any key to close this window . . .
```

Thank you